

Integrated Home Lighting System
Sunlight Driven Blinds and Motion Controlled LEDs

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I. INTRODUCTION

Founded in 2026, BrightMind Engineering is a group of engineering students interested in creating practical, energy-efficient solutions for everyday use. Contributions to this report can be found in Appendix E, Table 3. The team has created an automatic blinds system that will be set up in a room. The system will turn the lights on and off automatically when the room is occupied and vacated, respectively. A motion sensor will be used to determine if people are in the room. This sensor will control the lights. Additionally, a photosensor will be placed by the window to determine the level of sunlight in the room. If it is bright, the DC motor will spin a shaft that will roll down the blinds. When it gets darker outside, the blinds will roll back up. If the blinds are rolled down and the motion sensor detects motion, the LED strip will turn on. For a full work flow of the system, see Appendix D, Figure 5.

The design emphasizes reliability, efficiency, operability, and safety. The system is intended to require minimal maintenance, operate only when needed to conserve energy, and be easy for users to install and use. Safety considerations include proper mounting to prevent falling hazards and a safe electrical setup to avoid shock. Limitations include sensor accuracy, mechanical wear from repeated motor use, and changing environmental conditions.

This product addresses the serious problem of energy waste. “uncontrolled daylight can have negative impacts on the building environment. Intense daylight leads to glare problems at the perimeter zone and daylight with excessive solar gains leads to the increase of the cooling energy consumption” [1]. The problems of glare and unnecessary HVAC overuse show the need for such a product.

II. DESIGN SPECIFICATIONS

The design is a home automated lighting control system which will read the light levels outside, decide to close the blinds, and then, when motion is detected in the vicinity, will turn on lights.

The first subsystem is the photoresistor (see Appendix B, Fig. 3); this is used to detect the brightness of the sunlight. This subsystem runs all the other systems. The photoresistor was chosen instead of a photodiode or a phototransistor to limit unnecessary additional cost. Next is the motor (see Appendix B, Fig. 3); this is used to roll the blinds up or down when the brightness of the sunlight hits a certain threshold. When the brightness is less than 200 ADC, it rolls the blinds up if they are down, and if it is above 1000 ADC, it rolls the blinds down. The position of the motor/blinds is what runs the motion sensor and thus the LED lights. The basic motor was chosen to avoid additional costs and required less wiring and coding. The final subsystem is the motion sensor (see Appendix B, Fig. 3). When the brightness is beyond the threshold and the motor has turned the blinds down, the motion sensor is activated to detect motion. If all these conditions are met and the motion sensor detects motion, it turns on the LED lights to illuminate the room. One solution was to adjust the time and distance dials on the motion sensor to avoid frequent light changes and motion detection. This allowed the lights to stay on for about 2 minutes when motion is detected and detect distances up to 3 meters instead of 7 meters.

The electrical engineering analysis evaluates the voltage, current, and power consumption of the entire integrated circuit, the details of which can be found in Appendix B, Figure 4 and Table 1. In summary, the total power used by the circuit comes out to roughly 17.5 W. This assumes, however, that the motor is running and the lights are on at this moment. In reality, the motor will only be running for an estimated ten seconds per day, as it rolls down the blinds for

five seconds in the morning and raises them for five seconds at night. Thus, the power consumption will be around 15 W only when the LEDs are on, which is dependent on motion in the room. Thus, the total energy used will be at most around 60 Wh per day, assuming usage for four hours. Comparing this to the United States average of 751 Wh per day per bedroom lighting, this reduces energy consumption by 92% [2].

The sensor model uses the HC-SR501 PIR infrared sensor. This sensor detects radiation emitted from humans or animals and can be calibrated to detect motion at different sensitivities and angles [3]. The team was able to calibrate it to specific needs to detect motion within a distance of three meters from the sensor. A photoresistor was also used to determine if there is significant light outside to lower the blinds.

The actuation process uses an L293 H-bridge to run a DC motor so that it can spin both clockwise and counterclockwise. For the mechanics of the project, the motor is coupled with the light dowel rod, which is attached to the curtains. The motor spins to provide torque at a rotational speed to lower and raise the blinds. A housing was printed for the Arduino Board to improve the safety and aesthetics of the final product. The details of which can be found in Appendix A Figures 1 and 2.

From preliminary research, the team was able to identify accepted values of resistance based on light for the photoresistor:

The resistance of the photoresistor changes with the intensity of the incident light (visible light). Under dark conditions, its resistance (dark resistance) can reach 1~10 M Ω ; under strong light conditions (100 LX), its resistance (bright resistance) is only a few hundred to thousands of ohms. The sensitivity of the photoresistor to light (the spectral characteristics) is very close to the human eye's

response to visible light (0.4~0.76 μm). As long as the human eye can sense the light, it will cause its resistance to change. [4]

From experimentation with the photoresistor, shining a phone light of around 30 lumens gives an approximate resistance of 50 Ω . Covering the photoresistor with a hand gives an approximate resistance of 20 k Ω .

When testing with the Arduino, ambient light from the room provided an ADC value of roughly 960. When shining a phone light on it, a value of 1010–1020 was displayed. When a dark covering was put on it, a reading of 50 was shown.

To add actuation to the code, first the code runs with the room light. The motor did not spin. Then, a bright light from a phone shined on it, and the motor spun clockwise for 5 seconds and then turned off. This simulated the blinds rolling down when it was very bright outside. Next, the blinds sat for a minute and nothing happened, as the blinds should stay down for the rest of the day. Then, when it gets dark outside (simulated by putting a dark covering over the photoresistor), the motor spins counterclockwise for 5 seconds to roll the blinds back up. The final code can be seen in Appendix D.

Overall, this project was moderately expensive to build (as seen in Appendix C, Table 2). The overall cost to build this project was \$79.11. If this device had 1000 copies mass-produced, it would cost \$63,288; this includes a 20% price reduction due to components being bought in bulk.

III. TECHNICAL ISSUES

There were several major issues faced with this project. One of the main issues was getting the LED strip to fully light up. When powering it from the Arduino, it was only able to get roughly 7 of the lights to actually light up when supplied with 5 volts. Then, testing was done to see if it would work with 12 volts, and the whole strip was brightly lit for a minute and then turned off. The LED strip had to be reordered. This made the team realize that the power had to come from an external source of 5 volts, as the Arduino does not provide nearly enough current to power all 60 lights. Another issue that was encountered was with the motion sensor. During testing, the motion sensor would always detect motion even when it was covered. This was because the pins were connected in the wrong order, which caused this malfunction. The pins were then properly arranged to allow the motion sensor to correctly detect motion.

If the team could redesign the project, it would prioritize utilizing more of the budget. Another consideration was to purchase a more powerful motor so that the curtain rod would move more quickly and smoothly. Redesigning the circuit box would also be a priority, improving it to make it smaller and more compact. If there were more time, the LED strip color could be programmable so that the user could change it to their desired color.

IV. CONCLUSION

This product offers a convenient and energy efficient way to control both blinds and lighting systems. By combining motion and light sensing, it reduces unnecessary energy use. It is easy to use and requires minimal maintenance.

Compared to existing products, this system combines multiple functions into one system, rather than requiring separate devices for lighting and blinds. This system adapts to real-world conditions, which makes it more versatile than standard automated systems.

V. REFERENCES

1. Koo, S. Y., Yeo, M. S., & Kim, K. W. (2010). *Automated blind control to maximize the benefits of daylight in buildings*. Building and Environment, 45(6), 1508–1520.
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https://www1.eere.energy.gov/buildings/publications/pdfs/ssl/2012_residential-lighting-study.pdf
3. Amuta, E., et al. (2024). *[Title of article]*. IOP Conference Series: Earth and Environmental Science, 1342(1), 012001.
<https://doi.org/10.1088/1755-1315/1342/1/012001>
4. Kynix. (n.d.). *Photoresistor basics and Arduino tutorial*. Kynix.
<https://www.kynix.com/Blog/photoresistor-basics-arduino-tutorial.html>

APPENDIX A. MECHANICAL DESIGN DRAWINGS.

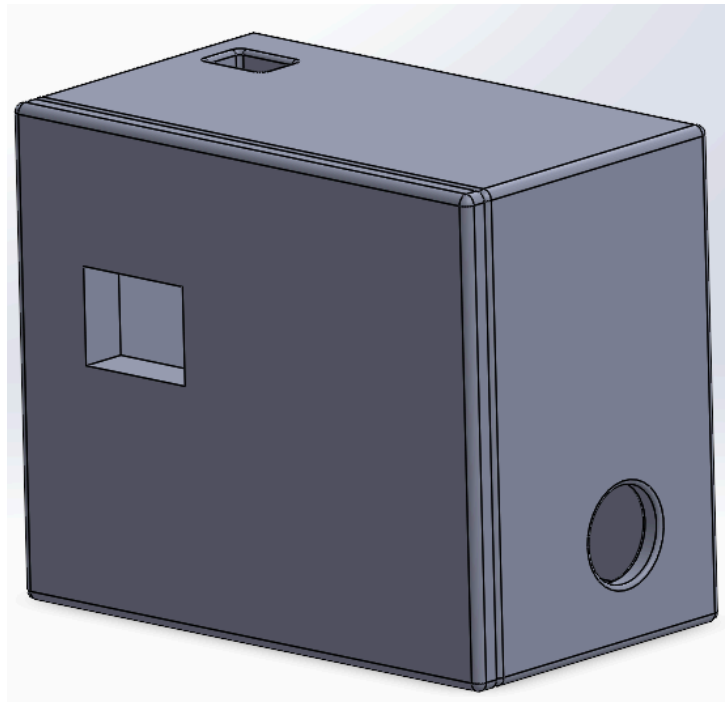


Figure 1. SolidWorks Drawings of Arduino Housing.

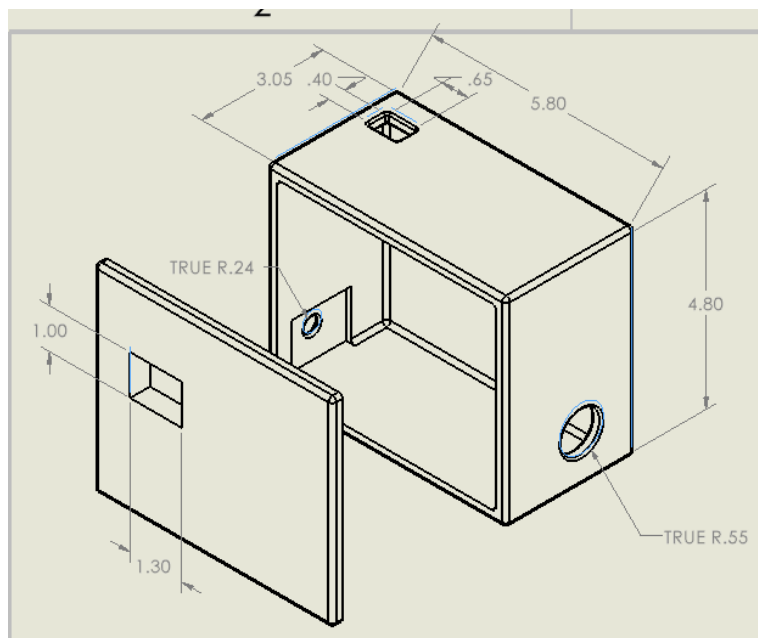


Figure 2. SolidWorks Parts Assembly of Arduino Housing.

APPENDIX B. ELECTRICAL SCHEMATICS.

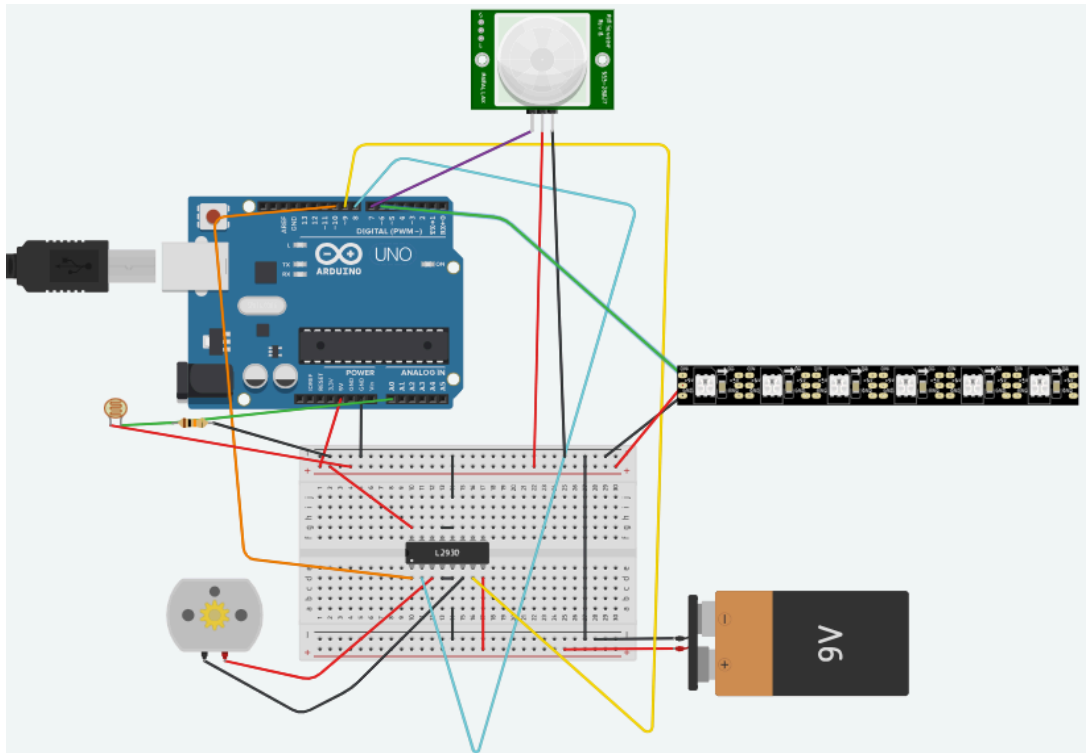


Figure 3. TinkerCad Schematic of the integrated circuit. All three subsystems as described in Section II are represented in the diagram above. The photosensor subsystem is integrated as a voltage divider, connecting to the motor subsystem through the H-bridge. The Motion sensor and lights subsystems are also interconnected, as shown above.

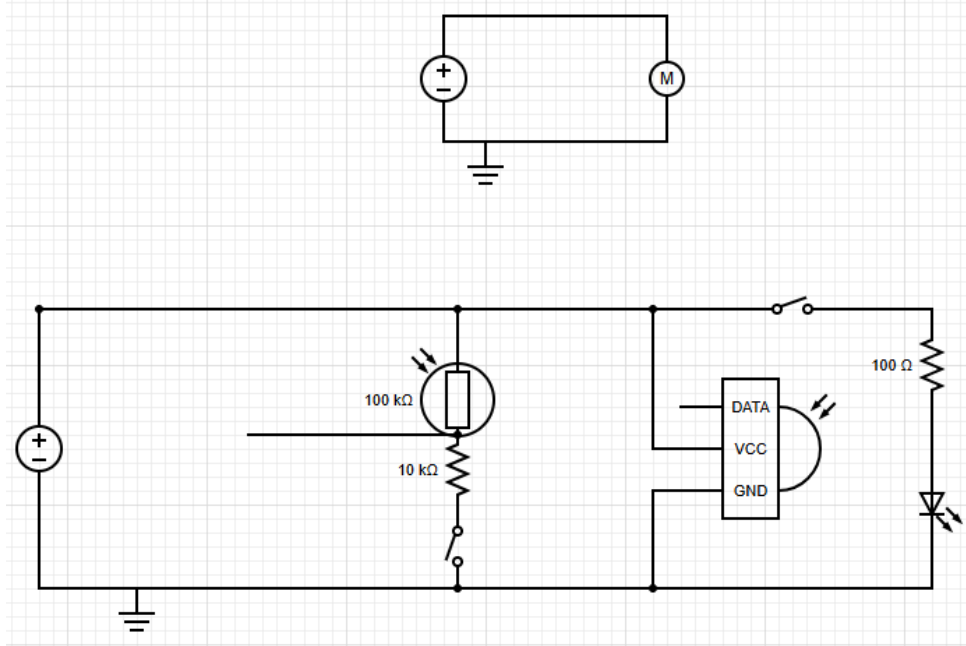


Figure 4. Simplified Electrical Drawings of circuit for analysis

Table 1. Summary of Electrical Analysis.

Component	Voltage (V)	Current (A)	Power (W)
LDR (Photoresistor)	4.545	0.0000455	0.000206797
10 kΩ Resistor (Divider)	0.455	0.0000455	2.07025E-05
LDR Branch Total	5	0.0000455	0.0002275
PIR Motion Sensor	5	0.065	0.325
LED Strip (60 LEDs, max)	5	2.88	14.4
Main 5V Circuit Total	5	2.945	14.725
DC Motor (estimated)	9	0.3	2.7

APPENDIX C. COST AND MATERIALS.

Table 2. Cost Analysis and Bill of Materials for the integrated system.

Item #	Component	Description	Quantity	Unit Cost (\$)	Total Cost (\$)	Notes
1	Motion Sensor	PIR motion sensor	1	\$3.50	\$3.50	Purchased
2	LED Strip	WS2812B addressable RGB LED strip (3.28 ft, 60 LED/m)	1	8.99	8.99	Purchased
3	Curtain Rod	1" x 36" Birch wood dowel	1	9.05	9.05	Purchased
4	Rod Holder	Adjustable curtain rod brackets	1	6.99	6.99	Purchased
5	Outer Shell	Custom 3D printed housing	1	2.58	3.58	Fabricated

6	Photoresistor	Light sensor	1	1.40	1.40	Provided
7	DC Motor	Small DC motor	1	10.00	10.00	Provided
8	L293D H-Bridge	Motor driver IC	1	2.63	2.63	Provided
9	Arduino	Microcontroller board	1	9.99	9.99	Provided
10	Wiring	Jumper wires	Assorted	0	0	Provided
11	Power Supply	Battery or DC supply	1	7.99	7.99	Provided
12	Curtains and Plywood	Fabric curtains and plywood	1	0	14.99	Provided
				Total Cost	79.11	

APPENDIX D. PROGRAMMING CODE AND WORKFLOW.

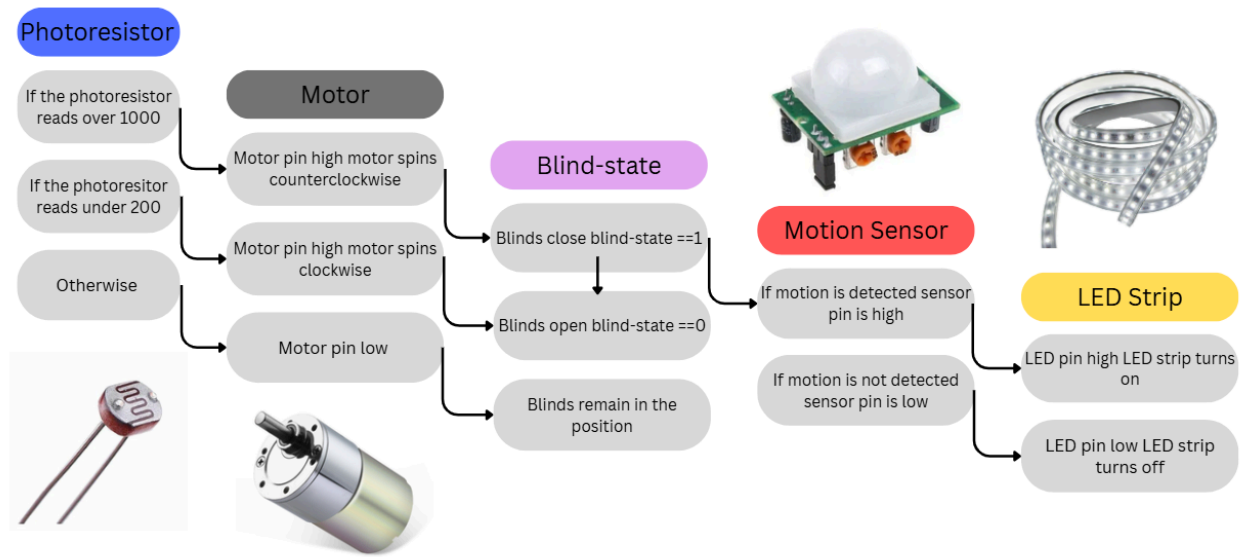


Figure 5- Flow chart of the system.

Programing Code Uploaded to Arduino:

```
#include <Adafruit_NeoPixel.h> // LED library
int Sensor_PIN = A0;
int INPUT1 = 8;
int INPUT2 = 9;
int ENABLE = 10;
int BlindState = 0;
int SensorValue = 0;

int MOTION_PIN = 7;
int LED_PIN = 6;

int threshold = 200; // Blinds roll back up below threshold

#define NUM_LEDS 60

Adafruit_NeoPixel strip(NUM_LEDS, LED_PIN, NEO_GRB + NEO_KHZ800); //LED setup
```

```

void setup() { // Declare all of the pins used.
  pinMode(Sensor_PIN, INPUT);
  pinMode(INPUT1, OUTPUT);
  pinMode(INPUT2, OUTPUT);
  pinMode(ENABLE, OUTPUT);
  pinMode(MOTION_PIN, INPUT);

  Serial.begin(9600);

  strip.begin();
  strip.show(); // all OFF

  digitalWrite(ENABLE, HIGH);

  delay(3000);
}

void loop() {

  // read Sensor
  SensorValue = analogRead(Sensor_PIN);
  Serial.println(SensorValue);           // Gives ADC converted brightness of photoresistor

  if (SensorValue > 1000 && BlindState == 0) {
    // Close blinds
    digitalWrite(ENABLE, HIGH);
    digitalWrite(INPUT1, LOW);
    digitalWrite(INPUT2, HIGH);

    delay(5000);

    digitalWrite(ENABLE, LOW);

    BlindState = 1;
    Serial.println(BlindState);
  }

  else if (SensorValue < threshold && BlindState == 1) {
    // Open blinds

```

```
digitalWrite(ENABLE, HIGH);
digitalWrite(INPUT1, HIGH);
digitalWrite(INPUT2, LOW);

delay(5000);

digitalWrite(ENABLE, LOW);

BlindState = 0;
}

else {
    digitalWrite(ENABLE, LOW);
}

// Motion detection with LED strip
if (BlindState == 1) {
    int motion = digitalRead(MOTION_PIN);

    Serial.print("Motion: ");
    Serial.println(motion);

    if (motion == HIGH) {
        // Warm white-ish color
        for (int i = 0; i < NUM_LEDS; i++) {
            strip.setPixelColor(i, strip.Color(255, 180, 100));
        }
        strip.show();
    } else {
        strip.clear();
        strip.show();
    }
} else {
    strip.clear();
    strip.show();
}
}
```

APPENDIX E. CONTRIBUTIONS

Table 3. Report Contributions.

Section	Contributors
I. Introduction	Kearsten
II. Design Specifics	Lola, Tyler, Kearsten
III. Issues and Problems	Lola, Tyler
IV. Conclusions	Kearsten
Appendix A. SolidWorks	Brady
Appendix B. Electrical Schematics	Brady, Tyler
Appendix C. Materials and Cost	Tyler, Lola
Appendix D. Programming Codes	Tyler, Brady